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Specimen transport device, specimen analysis system, specimen pretreatment system, and specimen transport method

Abstract

Provided are a specimen transport device, a specimen analysis system, a specimen pretreatment system, and a specimen transport method that can reduce human errors as compared with the related art. There are a plurality of magnetic bodies having different inductance characteristics, and a calculation unit calculates inductance characteristic values at a predetermined current value flowing through a magnetic pole detected by a current detection unit and specifies the type of the magnetic body having different inductance characteristics based on the obtained characteristic values to identify a holder.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a specimen transport device used in a specimen analysis system for analyzing a biological sample (hereinafter, referred to as a specimen) such as blood, plasma, serum, urine, and other body fluid or a specimen pretreatment system for performing pretreatment necessary for analysis, and a specimen transport method.

BACKGROUND ART

(2) As an example of a laboratory sample distribution system and a corresponding method of operation that are highly-flexible and offer a high transport performance, PTL 1 discloses a laboratory sample distribution system including more than one tube carrier, each of the tube carriers including at least one magnetically active device, preferably including at least one permanent magnet, and being adapted to transporting a sample tube, a transport plane adapted to transporting the tube carriers, and more than one electromagnetic actuators stationary below the transport plane, the electromagnetic actuators being adapted to moving the tube carriers on the transport plane by applying a magnetic force to the tube carriers.

(3) As an example of a laboratory sample distribution system in which a position on a transport surface can be recognized, PTL 2 describes a laboratory sample distribution system including a transport surface, a plurality of sample tube carriers, a drive unit configured to move the sample tube carriers on the transport surface, and a control device configured to control the movement of the sample tube carriers on the transport surface by driving the drive unit such that the sample tube carriers move along a corresponding transport path, in which a plurality of optically recognizable geometric shapes are placed on the transport surface, and each of the geometric shapes represents a dedicated field on the transport surface.

CITATION LIST

Patent Literature

(4) PTL 1: JP-A-2017-77971 PTL 2: JP-A-2018-119962

SUMMARY OF INVENTION

Technical Problem

(5) Examples of a specimen processing system for automatically analyzing a specimen include a specimen pretreatment system that performs input, centrifugal separation, dispensing processing, labeling processing and the like of the specimen, and a specimen analysis system that analyzes the

specimen processed by the specimen pretreatment system.

(6) In the specimen pretreatment system or the specimen analysis system, a transport line of a specimen including a belt conveyor or the like is provided to transport the specimen to a mechanism that performs predetermined processing or analysis. The specimen is transported to a predetermined mechanism by mounting a plurality of the transport lines on a specimen transport device.

(7) In recent years, an importance of specimen processing has been increased due to sophistication of medical treatment and a progress of an aging society. Therefore, in order to improve an analysis processing capacity of the specimen analysis system, high-speed transport, simultaneous mass transport, and transport in a plurality of directions of a specimen are desired.

(8) Examples of a technique of implementing such transport, include techniques described in PTLs 1 and 2.

(9) Here, in a specimen transport device, various specimens such as a general specimen for performing normal processing, an emergency specimen for which rapid transport and analysis processing are required, or a control specimen for performance confirmation are transported and processed, and a user visually identifies a specimen tube according to use.

(10) Therefore, in a transport method by an electromagnetic actuator using specimen holders described in PTLs 1 and 2, a transport operation of a plurality of specimen holders is performed in a specimen transport unit on a plane, and a human error such as misplacement of the specimen to the specimen holders may occur. Accordingly, improvement is desired.

(11) An object of the invention is to provide a specimen transport device, a specimen analysis system, a specimen pretreatment system, and a specimen transport method capable of reducing human errors as compared with those in the related art.

Solution to Problem

(12) The invention includes a plurality of methods for solving the above problems, and as an example thereof, there is provided a specimen transport device including: a first magnetic body provided in a transport holder in which a specimen tube storing a specimen is installed; a magnetic circuit including a second magnetic body and a winding wound around the outer periphery of the second magnetic body; a drive unit configured to supply a current to the winding of the magnetic circuit; a current detection unit configured to detect a current value flowing in the magnetic circuit; and a calculation unit configured to identify the transport holder based on a predetermined current value flowing through the magnetic circuit detected by the current detection unit, in which the first magnetic body includes a plurality of magnetic bodies having different inductance characteristics, and the calculation unit calculates an inductance characteristic value at a predetermined current value flowing through the magnetic circuit detected by the current detection unit and specifies the type of the first magnetic body having the different inductance characteristics based on the obtained inductance characteristic value to identify the transport holder.

Advantageous Effect

(13) According to the invention, human errors can be reduced as compared with those in the related art. Problems, configurations, and effects other than those described above will be clarified with the following description of examples.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a plan view showing a configuration of the entire specimen analysis system including a specimen transport device according to Example 1 of the present invention.

(2) FIG. 2 is a configuration diagram of the specimen transport device according to Example 1.

(3) FIG. 3 is a current superimposition characteristic graph of the inductance of a magnetic body in

the specimen transport device according to Example 1.

(4) FIG. 4 is a flowchart illustrating an operation of the specimen transport device according to Example 1 at the time of detecting a holder.

(5) FIG. 5 is a configuration diagram of an example of a holder according to the present invention.

(6) FIG. 6 is a configuration diagram of a transport device according to Example 2 of the present invention.

(7) FIG. 7 is a configuration diagram of a transport device according to Example 3 of the present invention.

DESCRIPTION OF EMBODIMENTS

(8) Hereinafter, examples of a specimen transport device, a specimen analysis system, a specimen pretreatment system, and a specimen transport method of the invention will be described with reference to the drawings.

(9) In the following examples, it is needless to say that elements (including element steps and the like) are not necessarily essential unless otherwise particularly specified or clearly considered as essential in principle.

(10) In all the drawings for showing the examples, the same members are denoted by the same reference numerals in principle, and the repeated description thereof is basically omitted.

Example 1

(11) Example 1 of the specimen transport device, the specimen analysis system, the specimen pretreatment system, and the specimen transport method of the invention will be described with reference to FIGS. 1 to 5.

(12) First, a configuration of the entire specimen analysis system including the specimen transport device will be described with reference to FIG. 1. FIG. 1 is a plan view showing the configuration of the entire specimen analysis system including the specimen transport device according to Example 1 of the invention.

(13) In FIG. 1, a specimen analysis system **100** according to Example 1 is a system for automatically analyzing a component of a specimen such as blood or urine.

(14) Main components of the specimen analysis system **100** include a specimen input unit **101**, a specimen storage unit **102**, a centrifugal processing unit **103**, a plug removing processing unit **104**, a secondary specimen tube generation processing unit **105**, a dispensing processing unit **106**, a plug attaching processing unit **107**, analysis processing units **108**, a specimen transport unit **109**, and a control unit **110**.

(15) The specimen input unit **101** inputs a specimen tube **201** storing a specimen into the specimen analysis system **100**. The specimen input unit **101** includes a specimen recognition unit, a plug detection unit, and a specimen holder recognition unit (all of which are omitted for convenience of illustration). The specimen input unit **101** recognizes a tube type of the specimen tube **201** (see FIG. 2) to be transported, a shape of a plug of a tube, and ID information assigned to a holder **202** (see FIG. 2) in which the specimen tube **201** is disposed, and obtains information for specifying the specimen tube **201** to be transported.

(16) The specimen holder recognition unit is provided at plural positions in the specimen analysis system **100**, and a location of the specimen tube **201** can be confirmed by the specimen holder recognition unit at the positions.

(17) The centrifugal processing unit **103** performs centrifugal separation on the input specimen tube **201**.

(18) The plug removing processing unit **104** is a unit that performs a process of removing the plug from the input specimen tube **201**.

(19) The secondary specimen tube generation processing unit **105** prepares another specimen tube **201** necessary for dispensing the specimen stored in the input specimen tube **201** into the subsequent dispensing processing unit **106**, and attaches a bar code or the like.

(20) The dispensing processing unit **106** parcels out the specimen into the other specimen tube **201**

prepared by the secondary specimen tube generation processing unit **105** to analyze the uncentrifuged specimen or the specimen centrifuged by the centrifugal processing unit **103** by the analysis processing units **108** or the like.

(21) The plug attaching processing unit **107** is a unit that attaches the plug to the specimen tube **201** from which the plug was removed and the specimen tube **201** into which the specimens are parceled out. The specimen analysis system **100** may also include two or more plug attaching processing units **107** in accordance with a type of the plug used to be attached to the specimen tube **201**.

(22) The analysis processing units **108** transport the specimen processed by the processing units in the specimen analysis system **100** and perform qualitative and quantitative analysis of a component of the specimen. Analysis items in the units are not particularly limited, and a configuration of a known automatic analysis device for analyzing biochemical items and immune items may be employed. Further, when a plurality of analysis processing units are provided, the analysis processing units may have the same or different specifications, and are not particularly limited.

(23) The specimen storage unit **102** is a unit that stores the specimen tube **201** from which the plug has been removed by the plug attaching processing unit **107**.

(24) The specimen transport unit **109** transports the specimen tube **201** input from the specimen input unit **101** or the specimen tube **201** in which the specimen is dispensed by the dispensing processing unit **106** and parceled out to units in the specimen analysis system **100** such as the centrifugal processing unit **103**, the dispensing processing unit **106**, and the analysis processing units **108**. The specimen transport unit **109** is also used for transport to mechanisms that perform a predetermined operation in units such as the centrifugal processing unit **103**, the dispensing processing unit **106**, and the analysis processing unit **108**.

(25) Accordingly, the holder **202** in which the specimen tube **201** input from the specimen input unit **101** is installed is transported to a predetermined pretreatment unit via the specimen transport unit **109**, and then transported to the analysis processing units **108**.

(26) The control unit **110** controls the operation of units in the specimen analysis system **100** and mechanisms in the units and analyzes measurement data in the analysis processing units **108**, and is implemented by a computer including a display device such as a liquid crystal display, an input device, a storage device, a CPU, a memory, and the like. The control unit **110** can confirm the location of the specimen in the specimen analysis system **100** from the ID information of the holder **202** by communicating with the units and the mechanisms described above.

(27) Control of the operation of devices by the control unit **110** is executed based on various programs recorded in the storage device in the control unit **110**.

(28) Operation control processing executed by the control unit **110** may be integrated into one program, may be divided into a plurality of programs, or may be performed in a combination thereof. Further, a part or all of the programs may be implemented by a dedicated hardware, or may be modularized.

(29) In FIG. **1**, the specimen transport unit **109** is implemented by a combination of a plurality of specimen transport devices of the invention described later. Alternatively, the specimen transport device of the invention may be implemented by the specimen transport device described later, or a combination of a specimen transport device not including a calculation unit **210** described later and the specimen transport device described later. The specimen transport unit **109** may include at least one or more specimen transport devices of the invention.

(30) In FIG. **1**, the specimen analysis system **100** includes various types of pretreatment units such as the specimen input unit **101** for specimen pretreatment. Alternatively, the specimen analysis system may be used as a system (a system in which the plurality of analysis processing units **108** are connected by the specimen transport device) not including the units for pretreatment.

(31) The invention can also be applied to a transport device that connects units in a specimen pretreatment system in which the analysis processing units **108** are omitted from the specimen

analysis system **100**.

(32) Next, a configuration of the specimen transport device according to Example 1 of the invention will be described with reference to FIG. 2. FIG. 2 is a configuration diagram of the specimen transport device according to Example 1 of the invention.

(33) In FIG. 2, a plurality of holders **202** in each of which the specimen tube **201** storing the specimen is installed are provided in the specimen transport device. Magnetic bodies **203** having different inductance characteristics depending on a difference in at least one of the material, the density, and the shape are provided on a bottom surface portion of each of the plurality of holders **202**.

(34) The magnetic body **203** is implemented by a permanent magnet such as neodymium and ferrite, and may also be implemented by other magnets and soft magnetic bodies, and the materials can be appropriately combined.

(35) It is not necessary that all the inductance characteristics of the magnetic bodies **203** of the holders **202** in the specimen transport device are different, as long as the inductance characteristics are divided into at least two types of ranges including a group of inductance characteristics according to specifications within a specific range and a group of inductance characteristics according to specifications within a specific range distinctly different from the range in the other group.

(36) It is not necessary to provide the magnetic body **203** on a lower surface of the holder **202**, and it is desirable to provide the magnetic body **203** on the lower surface from a viewpoint of a range over which a transport force of electromagnetic transport is exerted.

(37) The holder **202** including the magnetic body **203** slides on a transport surface **204**. In order to generate the transport force, a plurality of magnetic poles **207** each including a columnar core **205** and a winding **206** wound around an outer periphery of the core **205** are provided below the transport surface **204**.

(38) The magnetic pole **207** is connected to a drive unit **208** that applies a predetermined voltage to the magnetic pole **207** to flow a predetermined current through the winding **206**. In the present example, the drive unit **208** is similarly connected to the other magnetic poles **207**, which is omitted for convenience of illustration.

(39) The magnetic pole **207** to which the voltage is applied by the drive unit **208** serves as an electromagnet, and attracts the magnetic body **203** in the holder **202** on the transport surface **204**. After the holder **202** is attracted by the magnetic pole **207**, voltage application from the drive unit **208** to the magnetic pole **207** is stopped, and the magnetic body **203** in the holder **202** is attracted to the adjacent magnetic pole by applying the voltage from the drive unit **208** to a different magnetic pole adjacent to the magnetic pole **207** in the same manner as described above.

(40) By repeating this procedure on all the adjacent magnetic poles **207**, the holder **202** moves on the transport surface **204**, that is, the specimen stored in the specimen tube **201** held in the holder **202** provided with the magnetic body **203** is transported to a destination.

(41) The current flowing through the winding **206** of the magnetic pole **207** during transport is detected by a current detection unit **209**. The current detection unit **209** executes a current detection step of detecting a current value flowing when the current is supplied to the winding **206** of the magnetic pole **207**. The current flowing through the winding **206** of the magnetic pole **207** detected by the current detection unit **209** is transmitted to the calculation unit **210** to be digitized.

(42) In the present example, it is desirable that the current detection unit **209** detects the current value when each of the different holders **202** is in the same region, more preferably when each of the different holders **202** is stopped in the same region. However, the current value may be detected during transport. Even during the transport, it is desirable to detect the current value when each of the different holders **202** is in the same region.

(43) As a unit for specifying the holders existing in the same range, a known unit such as a Hall element or various units can be used.

- (44) The calculation unit **210** calculates the current flowing through each winding **206** using various pieces of information such as position information, speed information, and weight information on the holder **202**, and outputs a command signal to the corresponding drive unit **208**. The drive unit **208** applies a voltage to the corresponding winding **206** based on the command signal.
- (45) In addition, the calculation unit **210** of the present example identifies the holder **202** based on a predetermined current value flowing through the magnetic pole **207** detected by the current detection unit **209**. More specifically, the holder **202** is identified by calculating inductance characteristic values **403A** and **403B** (see FIG. 3) at the predetermined current value flowing through the magnetic pole **207** detected by the current detection unit **209** and specifying a type of the magnetic body **203** having different inductance characteristics depending on the obtained inductance characteristic values **403A** and **403B**. That is, a calculation step is executed by the calculation unit **210**.
- (46) Hereinafter, a current superimposition characteristic of an inductance of the magnetic body **203** of the holder **202** according to the invention will be described with reference to FIG. 3. FIG. 3 is a graph showing the current superimposition characteristic of the inductance of the magnetic body **203** according to the present example.
- (47) The current superimposition characteristic is a characteristic that the inductance characteristic value decreases as the magnetic body approaches magnetic saturation when a direct current flows, and the inductance characteristic value decreases by increasing the flowing current.
- (48) The magnetic body **203** in the holder **202** has an inductance waveform **401A** or an inductance waveform **401B** indicating a specific current superimposition characteristic as shown in FIG. 3 depending on the material, the shape, and the like. The inductance characteristic value **403A**, which is a value of the inductance waveform **401A** at a specific current value **402**, and the inductance characteristic value **403B**, which is a value of the inductance waveform **401B**, are stored in the calculation unit **210**.
- (49) By setting a predetermined threshold **404** to a range in which the inductance characteristic values **403A** and **403B** can be identified from each other, a difference between the magnetic bodies **203** can be identified, and the holders **202** can be identified into a holder type A and a holder type B. By this method, a type of the holder **202** (for general specimen, emergency specimen, standard sample, accuracy control sample, or the like) is identified, or the type of the specimen tube **201** (general specimen, emergency specimen, standard sample, accuracy control sample, or the like) installed in the holder **202** is identified.
- (50) The calculation unit **210** can be implemented by controlling a computer including a CPU, a memory, an interface, and the like or a field-programmable gate array (FPGA) to read programs and execute the calculation. These programs are stored in an internal recording medium or an external recording medium (not shown) in each of the configurations, and are read and executed by the CPU.
- (51) The operation control processing may be integrated into one program, may be divided into a plurality of programs, or may be a combination thereof. Further, a part or all of the programs may be implemented by dedicated hardware, or may be modularized. Further, the various programs may be installed in a corresponding device from a program distribution server, the internal recording medium, or the external recording medium.
- (52) Further, the programs are not necessary to be independent of the drive unit **208** and the like, and two or more may be integrated and shared, and only the processing may be shared. In addition, at least a part of the configuration may be connected via a wired or wireless network.
- (53) Next, a holder detection operation of the specimen transport device when the specimen according to the present example is transported will be described with reference to FIG. 4. FIG. 4 is a flowchart showing an operation of the transport device according to Example 1 of the invention when detecting the holder.

(54) As shown in FIG. 4, first, the transport of the holder **202** holding the specimen tube **201** storing the specimen is started by the specimen transport unit **109** of the specimen analysis system **100** (step S301).

(55) During the specimen transport operation, the holder **202** is transported to the magnetic pole **207** that detects the current flowing through the winding **206** by the current detection unit **209** shown in FIG. 2 (step S302).

(56) After the holder **202** is transported to above the predetermined magnetic pole **207**, the predetermined voltage is applied to the magnetic pole **207** from the drive unit **208** (step S303).

(57) The current value flowing through the winding **206** of the magnetic pole **207** due to the predetermined voltage applied in step S303 is detected by the current detection unit **209** and transmitted to the calculation unit **210** (step S304).

(58) In the storage device of the calculation unit **210**, a relationship between the current value flowing through the winding **206** of the magnetic pole **207** and the inductance characteristic value of the magnetic body **203** of the holder **202** corresponding to the current value is recorded in advance, and the inductance characteristic value of the magnetic body **203** of the holder **202** transported to above the magnetic pole **207** is detected based on the current value detected by the current detection unit **209** (step S305).

(59) In addition, the calculation unit **210** determines whether the inductance characteristic value of the holder **202** including the magnetic body **203** exceeds the predetermined threshold based on the current value detected in step S304 (step S305).

(60) When it is determined in step S305 that the inductance characteristic value of the holder **202** exceeds the threshold, the holder **202** is classified into the “holder type A” (step S306). The holder **202** identified into the holder type A is transported to a predetermined holder standby position for the holder type A (step S307).

(61) On the other hand, when the calculation unit **210** determines in step S305 that the inductance characteristic value of the holder **202** does not exceed the predetermined threshold, the holder **202** is identified into the “holder type B” (step S309). The holder **202** identified into the holder type B is transported to a predetermined holder standby position for the holder type B (step S307).

(62) The identification of the holder **202** is completed by moving the holder **202** to the holder standby position for the holder type A or moving the holder **202** to the holder standby position for the holder type B. An operator installs the specimen tube **201** on the holder **202** and the normal transport operation is continued (step S308).

(63) In FIGS. 3 and 4, the magnetic body **203** has two types of inductance characteristic values including the type A and the type B. Alternatively, even when the holder types include three or more types, it is possible to operate the three or more types of holders in the specimen transport device by repeating the processing of determining the holder in the same manner.

(64) In this case, a plurality of thresholds for the inductance characteristic value corresponding to the magnetic body **203** in the holder **202** are stored in the calculation unit **210**, and in step S305 of FIG. 4, the holders **202** are identified into a plurality of holder types by performing determination regarding the plurality of thresholds.

(65) By repeating the processing shown in FIG. 4, the holder **202** identified by the inductance characteristic value stands by while being identified at a predetermined position. Accordingly, for example, when the operator installs the specimen tube **201** of the general specimen in the holder **202**, the holder **202** for the general specimen identified into the holder type A, the holder type B, or the like is transported from the predetermined standby position to the position where the specimen tube **201** storing the general specimen is installed in the holder **202**, so that it is possible to avoid the operator from erroneously installing the specimen tube **201** of the general specimen in a holder different from the holder **202** for the general specimen, as compared with that in the related art.

(66) Similarly, by identifying the holder **202**, for example, when the emergency specimen requiring rapid transport and analysis processing is transported, it is also possible to transport the emergency

specimen on a transport path dedicated to the holder **202** for the emergency specimen, which is different from the holder **202** for the general specimen and is also provided on the specimen transport unit **109**, and a more rapid transport operation can be performed.

(67) Next, a structure of the holder **202** will be described. FIG. 5 is a configuration diagram of an example of the holder according to the invention.

(68) In the invention, since the holder type is determined based on the inductance characteristic value of the magnetic body **203** such as the permanent magnet in the holder **202**, the structure of the holder **202** itself is not necessarily limited to the structure in which one specimen tube **201** is installed, and may be a structure in which a plurality of specimen tubes **201** can be installed.

(69) For example, a holder as shown in FIG. 2 in which one specimen tube **201** is installed or a holder **202A** as shown in FIG. 5 that includes two or more connected holders **202** in FIG. 2 may be used.

(70) According to the structure of the holder **202A**, two or more specimen tubes **201** can be installed with the same holder information in a measurement requiring a large number of specimens in a large number of analysis items and the like. Accordingly, compared to a case in the related art where the specimen tubes **201** are installed in separate holders **202** and operated individually, it is possible to reduce an operation of matching information of the separate holders **202** in the related art with information of the specimen tubes **201**, and it is possible to reduce malfunctions such as specimen loss and specimen mistaking as compared with that in the related art.

(71) In addition to the holders **202** and **202A** holding one specimen tube **201** or two specimen tubes **201**, a specimen rack holding three or more specimen tubes **201** may be used.

(72) Next, an effect of the present example will be described.

(73) The specimen transport device according to Example 1 of the invention described above includes the magnetic body **203** provided in the holder **202** in which the specimen tube **201** storing the specimen is installed, the magnetic pole **207** including the core **205** and the winding **206** wound around the outer periphery of the core **205**, the drive unit **208** configured to supply a current to the winding **206** of the magnetic pole **207**, the current detection unit **209** configured to detect the current value flowing through the magnetic pole **207**, and the calculation unit **210** configured to identify the holder **202** based on the predetermined current value flowing through the magnetic pole **207** detected by the current detection unit **209**. The magnetic body **203** includes a plurality of magnetic bodies having different inductance characteristics, and the calculation unit **210** calculates the inductance characteristic values **403A** and **403B** at the predetermined current value flowing through the magnetic pole **207** detected by the current detection unit **209** and specifies the type of the magnetic body **203** having the different inductance characteristics based on the obtained inductance characteristic values **403A** and **403B** to identify the holder **202**.

(74) Accordingly, in a transport method by an electromagnetic actuator, it is possible to determine the type of the holder **202** in which the specimen tube **201** is installed by using the magnetic body **203** on the holder **202** that is necessary for transport, and it is possible to increase a material for determining the type as compared with that in the related art. Therefore, as the device and the specimen transport device, the specimen holder type can be automatically determined with higher accuracy, and the identification work by a user can be reduced or eliminated. Therefore, since the holder **202** can be prevented from being transported to a wrong position, effects of further improving work efficiency of the user and further reducing human errors can be obtained.

(75) In particular, although one holder can achieve more flexible transport corresponding to operating conditions than five racks do, according to the invention, it can be said that the effect of reducing errors is improved as compared with that in the related art since methods for determining the type of the holder **202** are increased and the type of the holder **202** can be identified with higher accuracy than that in the related art.

(76) Further, from an actual operation aspect of the specimen pretreatment system **120** or the specimen analysis system **100**, when the specimen transport device capable of identifying the type

of the holder **202** described above is disposed immediately before a buffer or the specimen storage unit **102** in which an empty holder after the processing by the specimen pretreatment system **120** and analysis by the specimen analysis system **100** remains, the effect of identifying the holder types with higher accuracy can be further received.

(77) In addition, since the magnetic body **203** has different inductance characteristics depending on the difference in at least one of the material, the density, and the shape, the inductance characteristics can be easily changed on the magnetic body **203**, and accuracy of identifying the holder **202** can be achieved with a simpler configuration.

(78) Further, since the current detection unit **209** detects the current value when each of the different holders **202** is in the same region, in particular, detects the current value when each of the different holders **202** is stopped in the same region, accuracy of obtaining the inductance characteristic values **403A** and **403B** can be improved, the type of the holder **202** can be more accurately identified.

Example 2

(79) A specimen transport device, a specimen analysis system, a specimen pretreatment system, and a specimen transport method according to Example 2 of the invention will be described with reference to FIG. 6. FIG. 6 is a configuration diagram of the specimen transport device according to Example 2 of the invention.

(80) The specimen transport device of the present example shown in FIG. 6 includes the plurality of magnetic poles **207**, a drive unit **208A** configured to individually apply a voltage to each of the plurality of magnetic poles **207**, a plurality of current detection units **209** configured to detect the current values flowing through the windings **206** of the magnetic poles **207** due to the voltage applied by the drive unit **208A**, and a calculation unit **210A** configured to identify types of the holders **202** in the magnetic poles **207** based on the current values detected by the plurality of current detection units **209**.

(81) Other configurations and operations are substantially the same as the configurations and operations of the specimen transport device, the specimen analysis system, and the specimen pretreatment system and the specimen transport method of Example 1 described above, and details are omitted.

(82) Also in the specimen transport device, the specimen analysis system, the specimen pretreatment system, and the specimen transport method according to Example 2 of the invention, the effects substantially the same as those of the specimen transport device, the specimen analysis system, and the specimen pretreatment system and the specimen transport method of Example 1 described above can be obtained.

(83) In addition, since there are a plurality of locations where inductance characteristic values of the holders **202** are obtained by including the plurality of magnetic poles **207**, the plurality of current detection units **209**, at least one or more drive units **208**, and at least one or more calculation units **210**, timings at which the types of the holders **202** are identified can be increased, and a structure in which a transport error is less likely to occur can be obtained.

(84) In addition, since the structures of the magnetic poles **207** and the drive unit **208** necessary for the normal specimen transport operation are the same as those of Example 1, the holder **202** can be identified with higher accuracy even during the transport operation of the holder **202**. Therefore, for example, even in the holder **202** in the transport operation in which the predetermined transport operation is completed and the holder **202** is returned to a predetermined holder standby position or a position where the specimen tube **201** is additionally installed, the holder **202** can be identified during the transport operation of returning, which can contribute to shortening time for preparing transport operation.

(85) In the present example, the calculation unit **210A** and the drive unit **208A** may be plural.

Example 3

(86) A specimen transport device, a specimen analysis system, a specimen pretreatment system, and

a specimen transport method according to Example 3 of the invention will be described with reference to FIG. 7. FIG. 7 is a configuration diagram of the specimen transport device according to Example 3 of the invention.

(87) In the specimen transport device of the present example shown in FIG. 7, a specific magnetic pole **207B** of the specimen transport device of Example 2 is a circuit in which shapes of a core **205B** and a winding **206B** are different from those of the cores **205** and the windings **206** of the other magnetic poles **207**. At least one or more magnetic poles **207B** may be provided in the specimen transport device, and the number thereof is not particularly limited.

(88) The shape of either the core **205B** or the winding **206B** of the specific magnetic pole **207B** may be different from those of the cores **205** and the windings **206** of the other magnetic poles **207**. In addition, not only the shape but also one or more of the material, the density, and the shape may be different from those of the cores **205** and the windings **206** of the other magnetic poles **207**.

(89) In the calculation unit **210B** of the present example, a threshold of an inductance characteristic value is set for each of the adjacent magnetic poles **207** having different configurations. Then, the different inductance characteristics of the magnetic bodies **203** are calculated based on detection values from a current detection unit **209B** that detects the current flowing through the specific magnetic pole **207B** and the current detection units **209** that detect the current flowing through the other magnetic poles **207**. Accordingly, during the normal transport operation of the holders **202**, the inductance characteristic values detected when the holders **202** pass above the magnetic poles **207** and **207B** are compared between the adjacent magnetic poles **207** and **207B**.

(90) A configuration of a drive unit **208B** that applies a voltage to the specific magnetic pole **207B** and the other magnetic poles **207** is the same as that of the drive unit **208A** of Example 2.

(91) The other configurations and operations are substantially the same as the configurations and operations of the specimen transport device, the specimen analysis system, and the specimen pretreatment system and the specimen transport method of Example 1 described above, and details are omitted.

(92) Also in the specimen transport device, the specimen analysis system, the specimen pretreatment system, and the specimen transport method according to Example 3 of the invention, the effects substantially the same as those of the specimen transport device, the specimen analysis system, the specimen pretreatment system and the specimen transport method of Example 1 described above can be obtained.

(93) The magnetic pole **207B** has a circuit in which one or more of the material, the density, and the shape of at least one of the cores **205**, **205B** and the windings **206**, **206B** are different, and the calculation unit **210B** can identify the holder type in a more detailed range by calculating the different inductance characteristics of the magnetic body **203** detected by the magnetic poles **207** and **207B** having different configurations. It is possible to provide a position where only a specific holder identification that is not related to the transport operation is performed, and the position can be used when an operator manually identifies the holder **202** in normal use and installs the specimen tube **201**.

(94) Similarly to the configurations shown in FIGS. 2 and 6, it is needless to say that the different magnetic pole **207B** shown in FIG. 7 can also be used as the magnetic pole used for the normal transport operation.

(95) <Others>

(96) The invention is not limited to the above examples, and includes various modifications. The examples described above have been described in detail for easy understanding of the invention, and are not necessarily limited to those including all the configurations described above.

(97) Further, a part of a configuration of one example can be replaced with a configuration of another example, and the configuration of the other example can be added to the configuration of the one example. A part of a configuration of each example may be added or replaced with another configuration, and one configuration may be deleted from the part of the configuration.

REFERENCE SIGN LIST

(98) **100**: specimen analysis system **101**: specimen input unit **102**: specimen storage unit **103**: centrifugal processing unit **104**: plug removing processing unit **105**: child specimen vessel generation processing unit **106**: dispensing processing unit **107**: plug attaching processing unit **108**: analysis processing unit **109**: specimen transport unit **110**: control unit **120**: specimen pretreatment system **201**: specimen vessel **202**, **202A**: holder (transport container) **203**: magnetic body (first magnetic body) **204**: transport surface **205**, **205B**: core (second magnetic body) **206**, **206B**: winding **207**, **207B**: magnetic pole (magnetic circuit) **208**, **208A**, **208B**: drive unit **209**, **209B**: current detection unit **210**, **210A**, **210B**: calculation unit **401A**, **401B**: inductance waveform **402**: current value **403A**, **403B**: inductance characteristic value **404**: threshold

Claims

1. A specimen transport device comprising: a first magnetic body provided in a transport container in which a specimen vessel storing a specimen is installed; a magnetic circuit including a second magnetic body and a winding wound around the outer periphery of the second magnetic body; a drive unit configured to supply a current to the winding of the magnetic circuit; a current detection unit configured to detect a current value flowing in the magnetic circuit; and a calculation unit configured to identify the transport container based on a predetermined current value flowing through the magnetic circuit detected by the current detection unit, wherein the first magnetic body includes a plurality of magnetic bodies having different inductance characteristics, and the calculation unit calculates an inductance characteristic value at a predetermined current value flowing through the magnetic circuit detected by the current detection unit and specifies the type of the first magnetic body having the different inductance characteristics based on the obtained inductance characteristic value to identify the transport container.
2. The specimen transport device according to claim 1, wherein the first magnetic body has a different inductance characteristic due to a difference in at least one of material, density, and shape.
3. The specimen transport device according to claim 1, wherein a plurality of the magnetic circuits, a plurality of the current detection units, at least one or more of the drive units, and at least one or more of the calculation units are provided.
4. The specimen transport device according to claim 1, wherein the magnetic circuit has a circuit having a configuration in which at least one of the material, density, and shape of at least one of the second magnetic body and the winding is different, and the calculation unit calculates each of the different inductance characteristics of the first magnetic body detected by the magnetic circuit having a different configuration.
5. The specimen transport device according to claim 1, wherein the current detection unit detects the current value when each of the different transport containers is present in the same area.
6. The specimen transport device according to claim 5, wherein the current detection unit detects the current value when each of the different transport containers is stopped in the same area.
7. A specimen analysis system comprising the specimen transport device according to claim 1.
8. A specimen pretreatment system comprising the specimen transport device according to claim 1.
9. A method for transporting a specimen stored in a specimen vessel held by a transport container provided with a first magnetic body, wherein the first magnetic body includes a plurality of magnetic bodies having different inductance characteristics, the method comprising: a current detection step of detecting a current value flowing when a current is supplied to a second magnetic body and a winding of a magnetic circuit having the winding wound around the outer periphery of the second magnetic body; and a calculation step of calculating an inductance characteristic value at a predetermined current value flowing through the magnetic circuit detected in the current detection step and specifying the type of the first magnetic body having the different inductance

characteristics based on the obtained inductance characteristic value to identify the type of the transport container.
